PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effect of obesity treatment interventions in preschool children aged two to six years: a systematic review and meta-analysis
AUTHORS	Nordlund, Samantha; McPhee, Patrick; Gabarin, Ramy; Deacon, Charlotte; Mbuagbaw, Lawrence; Morrison, Katherine

VERSION 1 – REVIEW

REVIEWER	Jull, Andrew The University of Auckland, School of Nursing
REVIEW RETURNED	07-Sep-2021

GENERAL COMMENTS	Thank you for the opportunity to review this work. I have a single major concern about this review. It is understandable that the review uses a date limited search strategy given the existence of a prior Cochrane review. However, the first objective in the abstract states the review intention is "To summarize the literature on the impact of pediatric weight management interventions on health outcomes in preschool age children with overweight or obesity." Limiting the analysis to the four trials published since 2016 adding these trials to those also identified in the earlier 2016 review means that this review summarises an incomplete evidence base and thus the conclusion at the end of the abstract, namely "Educational interventions delivered to the parents of young children with obesity result in decline in BMI z score" may not be justified as it only summarises a few of the possible trials that could have been included if this was an updated summary of the evidence in its entirety. My more minor concerns are: 1. It is not clear whether the subgroup analyses were planned a priori or were as a consequence of exploring for the cause of the significant heterogeneity (which would be rather limited with just four trials). 2. I can understand the rationale for splitting the control group a three arm trial, but using the same effect size and SD for each group will have impact on the overall variability over what is in effect four trials rather than three. While there is no alternative unless the trial reported individual participant data for the 50 participants, it should be acknowledged as a limitation especially given so few trials were reported. 3. I find it odd that the finding for BMI z score finding was not downgraded for serious imprecision when the 95%CI ranged from a significant effect to a non-significant effect, especially when another comparison is downgraded for the reason of having a wide confidence interval that crosses the line of no effect. Further I am not convinced that not downgrading the same comparison for inconsistency is justified - it is not clear that the causes of

	heterogeneity were explored and explained (hard, I acknowledge, in just 4 studies, which is another reason why at least some of 7 earlier studies from the Cochrane review should have been included in the analysis to better improve exploration of heterogeneity) - if they were not, then the subgroup analyses resolving heterogeneity could easily be a chance finding rather than actually resolving the heterogeneity. 4. The citation for the Cochrane review is incomplete as are a number of other citations eg 9, 12, 14, 24, 26. 5. Either the number of records screened (14018) or the number excluded (13576) or the number fullest screened (441) is incorrect bc 14018 - 13576 = 442.

REVIEWER	Kowal, Malgorzata Univ Sch Phys Educ Krakow
REVIEW RETURNED	14-Sep-2021

GENERAL COMMENTS	The topic of the effect of obesity treatment interventions in preschool children is interesting, especially now, when in most countries of the world there is a sharp increase in the prevalence of overweight and obesity among children, adolescents and adults. Nevertheless, in the current version, the text has some flaws and raises some questions: -first of all, it should be noted that the reviewed article refers to the Cochrane review and is largely similar to it in terms of structure. And it may raise some reservation of the Authors of this report - The Cochrane Report is extensive, approximately 80 pages long. The reviewed article is also extensive. However, the question arises whether such a volume of the review article is acceptable in the journal - it is doubtful that only 4 out of more than 14,000 publications were selected for meta-analysis. Which resulted in such a small number of publications used by the authors of the meta-analysis. Is it not too strict a selection? If so, then perhaps the articles used are not representative of the solution to the research question posed - in the materials and methods chapter the description of the publication qualification procedure used in the meta-analysis, it is extremely boring and difficult for the reader to analyze. It should be shortened and simplified - description of the search results, line 190-195 - it should be rather placed in the Material and Methods section - line 207 - 216. The authors of the study describe no change in BMI between the control group and the intervention group. It should be explained here what type of intervention the study group was given. Was it exercise, diet or encouragement to change your lifestyle Best regards

REVIEWER	Morris, Julie South Manchester NHS Trust, Medical Statistics Department
REVIEW RETURNED	22-Oct-2021

GENERAL COMMENTS	This systematic review assesses pediatric weight management interventions on BMI changes in preschool children. The differences between the protocol and the actual study systematic review with regards selection of studies and analyses

	of results are of concern. In addition, the presentation of results needs to be addressed: 1. The review includes RCTs only. However, the protocol states that observational studies will also be considered. Why were observational studies omitted? 2. Some details of the four RCTs included in the study are provided in Table 1, but more information should be given, either in the Table or text. No qualitative synthesis (as referenced in Figure 1) is carried out, and the description of the interventions is sketchy. For example, what is the 'no booster' arm in the Ek study? 3. The included studies show significant heterogeneity (as acknowledged on Page 11 in the Effect of interventions section). However, the meta-analysis results are still reported even though the protocol states that, "If there is substantial heterogeneity, a qualitative narrative summary will be done as opposed to a meta-analysis". 4. Subgroup analyses are carried out by intervention type (education vs MI). However, the protocol only mentions subgroup analyses for male vs female, overweight vs obese at baseline, and intervention setting (tertiary centre vs other). 5. A valuable discussion of the limitations of the systematic review is included in the main text. However, the effect of these limitations on the interpretation of the meta-analysis results should be included in the Abstract.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Prof. Andrew Jull, The University of Auckland, The University of Auckland

Comments to the Author:

Thank you for the opportunity to review this work. I have a single major concern about this review. It is understandable that the review uses a date limited search strategy given the existence of a prior Cochrane review. However, the first objective in the abstract states the review intention is "To summarize the literature on the impact of pediatric weight management interventions on health outcomes in preschool age children with overweight or obesity." Limiting the analysis to the four trials published since 2016 adding these trials to those also identified in the earlier 2016 review means that this review summarises an incomplete evidence base and thus the conclusion at the end of the abstract, namely "Educational interventions delivered to the parents of young children with obesity result in decline in BMI z score" may not be justified as it only summarises a few of the possible trials that could have been included if this was an updated summary of the evidence in its entirety.

Response: We thank the Reviewer for this important comment. As per your comment and a similar comment from the Editor, we have included evidence from the Cochrane review such that we now consider the evidence in its entirety. We include a specific section in the results: "Current study versus previous Cochrane systematic review" that explains the inclusion of evidence from the Cochrane

review. Additionally, we included an updated search (Nov. 21 2021) to identify any additional studies since the initial search.

My more minor concerns are:
1. It is not clear whether the subgroup analyses were planned a priori or were as a consequence of exploring for the cause of the significant heterogeneity (which would be rather limited with just four trials).

Response: We have added the following information to the Methods (synthesis of results) section (P.9 lines 182-188): If there was substantial heterogeneity, subgroup analyses based on sex or differences in BMI z score at baseline (e.g., overweight vs. obesity) were to be performed (PROSPERO CRD42020166843). We anticipated intervention modality (e.g., motivational interviewing or multicomponent educational interventions) would introduce heterogeneity and planned to conduct exploratory subgroup analyses with available data. If substantial heterogeneity remained, a qualitative synthesis was conducted.

2. I can understand the rationale for splitting the control group a three arm trial, but using the same effect size and SD for each group will have impact on the overall variability over what is in effect four trials rather than three. While there is no alternative unless the trial reported individual participant data for the 50 participants, it should be acknowledged as a limitation especially given so few trials were reported.

Response: We appreciate the expertise of the Reviewer considering the impact of splitting the control group of a three arm trial on overall variability. The Cochrane Handbook (5.1) recommends that multi groups or multicomponent interventions split the shared group (in this case the control arm) into two or more groups with smaller sample size, and to include two independent comparisons. As per the Reviewer's suggestion, we have added the following as a limitation (P.16 lines 364-366): Splitting the control group in three arm trials limits the overall variability within trials; however, it is a recommended approach in the Cochrane Handbook to analyze such multicomponent interventions.

3. I find it odd that the finding for BMI z score finding was not downgraded for serious imprecision when the 95%CI ranged from a significant effect to a non-significant effect, especially when another comparison is downgraded for the reason of having a wide confidence interval that crosses the line of no effect. Further I am not convinced that not downgrading the same comparison for inconsistency is justified - it is not clear that the causes of heterogeneity were explored and explained (hard, I acknowledge, in just 4 studies, which is another reason why at least some of 7 earlier studies from the Cochrane review should have been included in the analysis to better improve exploration of heterogeneity) - if they were not, then the subgroup analyses resolving heterogeneity could easily be a chance finding rather than actually resolving the heterogeneity.

Response: We agree with the Reviewer that it is important to explore the reasons for [statistical] heterogeneity. We attempted to further investigate the clinical heterogeneity by exploring subgroup analyses based on intervention modality (i.e., motivational interviewing vs. educational interventions). However, in response to concerns from the Reviewer and Editor to consider the entirety of the evidence, we have included an additional 4 trials from the Cochrane review (that met our inclusion criteria) plus an additional trial that was identified in our updated search. As such, we re-ran our meta-analyses and noted the following: A significant small effect for interventions resulted in a greater decline in BMI z-score compared to controls (MD -0.10, 95% CI -0.12 to -0.09; 8 trials, 1491 participants; $p<0.001$; I^2 68%).
Our protocol stated that subgroup analyses will be performed based on sex or differences in BMI z score at baseline, should the data permit. Unfortunately, this information was not provided in

individual trials; thus, we explored subgroup analysis based on intervention modality (motivational interviewing vs. educational intervention).

4. The citation for the Cochrane review is incomplete as are a number of other citations eg 9, 12, 14, 24, 26.

Response: We apologize for this oversight and have updated the citations accordingly.

5. Either the number of records screened (14018) or the number excluded (13576) or the number fullest screened (441) is incorrect bc 14018 - 13576 = 442.

Response: We apologize for this error and provide an updated PRISMA Flow Diagram including results from our initial and updated searches.

Reviewer: 2

Dr. Malgorzata Kowal, Univ Sch Phys Educ Krakow

Comments to the Author:

The topic of the effect of obesity treatment interventions in preschool children is interesting, especially now, when in most countries of the world there is a sharp increase in the prevalence of overweight and obesity among children, adolescents and adults. Nevertheless, in the current version, the text has some flaws and raises some questions.

Response: We thank the Reviewer for their comment. We hope that we have adequately addressed your questions below, and by doing so, have strengthened the quality of our manuscript.

1. First of all, it should be noted that the reviewed article refers to the Cochrane review and is largely similar to it in terms of structure. And it may raise some reservation of the Authors of this report.

Response: We acknowledge that the present article does have similar components to the Cochrane review in terms of structure. This is because we are using the recommended headings of Cochrane reviews as there is no formal layout specifications for this Journal.

2. The Cochrane Report is extensive, approximately 80 pages long. The reviewed article is also extensive. However, the question arises whether such a volume of the review article is acceptable in the journal.

Response: We appreciate the Reviewer's concern as an extensive article will impact the paper's 'readability'. We note that we are following BMJ Open's recommendation that the article should try not to exceed 4000 words.

3. It is doubtful that only 4 out of more than 14,000 publications were selected for meta-analysis. Which resulted in such a small number of publications used by the authors of the meta-analysis. Is it not too strict a selection? If so, then perhaps the articles used are not representative of the solution to the research question posed.

Response: The results in the present study closely align with other systematic reviews; particularly the Cochrane review, which included 7 RCTs from ~14,000 hits from database searches. The limited

evidence in preschooler obesity treatment is indeed identified from both the present review and the previous Cochrane review.

4. In the materials and methods chapter, the description of the publication qualification procedure used in the meta-analysis, it is extremely boring and difficult for the reader to analyze. It should be shortened and simplified.

Response: As per the Reviewer's suggestion, we have shortened the statistical analysis section such that it now includes only information pertinent to the present study.

5. Description of the search results, line 190-195 - it should be rather placed in the Material and Methods section

Response: We have placed information regarding database searches into the methods section such that Figure 1 is defined earlier in the manuscript. However, we elected to keep the results of full text screening under the "Search results" section of the manuscript.

6. Line 207 - 216. The authors of the study describe no change in BMI between the control group and the intervention group. It should be explained here what type of intervention the study group was given. Was it exercise, diet or encouragement to change your lifestyle.

Response: We have updated the results based on the inclusion of new trials from the Cochrane review (n=4) and updated search (n=1). We report these results on P.10 lines (223-227): There was a difference in BMI z score between intervention and control groups at the end of intervention (mean difference (MD) –0.10, 95% CI -0.12 to -0.09; 8 trials, 1491 participants; $p<0.0001$; I^2 68%) (Figure 2). No between group difference in %BMIp95 was identified (MD -1.34, 95% CI -3.17 to 0.49; 4 trials, 349 participants; $p=0.15$; I^2 79%) (Figure 3).

Reviewer: 3

Ms. Julie Morris, South Manchester NHS Trust

Comments to the Author:

This systematic review assesses pediatric/span> weight management interventions on BMI changes in preschool children.

The differences between the protocol and the actual study systematic review with regards selection of studies and analyses of results are of concern. In addition, the presentation of results needs to be addressed:

1. The review includes RCTs only. However, the protocol states that observational studies will also be considered. Why were observational studies omitted?

Response: Our study team decided that we would pursue observational studies if no RCT evidence was found. Given the quality of the evidence and overall effectiveness from the synthesized evidence, we added the following in the "Future directions" section (P.17 lines 387-389): …"emphasis on a continued search for novel pathways to intervene upon or new approaches to current paradigms must be undertaken given the modest outcomes to date, while also considering the relevance and potential impact of observational evidence".

2. Some details of the four RCTs included in the study are provided in Table 1, but more information should be given, either in the Table or text. No qualitative synthesis (as referenced in Figure 1) is carried out, and the description of the interventions is sketchy. For example, what is the 'no booster' arm in the Ek study?

Response: We would like to draw the Reviewer to the section in the results "Summary of included studies" where we provide a qualitative synthesis of the included trials. We have clarified "booster" in Table 1.

3. The included studies show significant heterogeneity (as acknowledged on Page 11 in the Effect of interventions section). However, the meta-analysis results are still reported even though the protocol states that, "If there is substantial heterogeneity, a qualitative narrative summary will be done as opposed to a meta-analysis".

Response: We have clarified the following in the Methods section (Synthesis of results) (P.9 lines 184-187): We anticipated intervention modality (e.g., motivational interviewing or educational interventions) would introduce heterogeneity and planned to conduct exploratory subgroup analyses with available data.

4. Subgroup analyses are carried out by intervention type (education vs MI). However, the protocol only mentions subgroup analyses for male vs female, overweight vs obese at baseline, and intervention setting (tertiary centre vs other).

Response: We acknowledge that we did not include intervention type in our protocol as a possible characteristic that might introduce clinic heterogeneity. In an effort to better understand the heterogeneity we saw in our analysis, we conducted an exploratory subgroup analysis (as noted above we were unable to conduct our a priori sub-group analyses due to insufficient data in the manuscripts) to examine intervention modality: motivational interviewing vs educational interventions. Thus, we conducted exploratory subgroup analyses based on intervention modality. This information is now included in the methods under "synthesis of results". As noted this had little influence in reducing heterogeneity.

5. A valuable discussion of the limitations of the systematic review is included in the main text. However, the effect of these limitations on the interpretation of the meta-analysis results should be included in the Abstract.

Response: We have added the following information to the abstract (conclusions): "The results should be interpreted cautiously as the quality of the evidence ranged from low to moderate."

VERSION 2 – REVIEW

REVIEWER	Kowal, Malgorzata Univ Sch Phys Educ Krakow
REVIEW RETURNED	06-Feb-2022

GENERAL COMMENTS	After the corrections, which were made by the Authors of manuscripte, I have no comments and recommend the article for publication.

REVIEWER	Morris, Julie South Manchester NHS Trust, Medical Statistics Department
REVIEW RETURNED	27-Jan-2022

GENERAL COMMENTS	This systematic review has been revised to incorporate an extended search period and has increased to nine the number of trials included in the study. However, concerns remain relating to the presentation of meta-analysis results in the presence of significant heterogeneity. 1. The results of a meta-analysis of all 9 included studies are reported in the Results section and in the Abstract, even though 'substantial heterogeneity' was present. The extent of such heterogeneity means that it is really inappropriate to present the summary effect size from the meta-analysis. In fact, the protocol (PROSPERO CRD42020166843) appears to be of the same opinion as it states that, "If there is substantial heterogeneity, a qualitative narrative summary will be done as opposed to a meta-analysis". 2. In terms of subgroup analysis, the protocol states that, "The following characteristics are thought to introduce clinical heterogeneity, and thus the following subgroup analyses are planned should the data permit: variation in sex (males vs female), differences in BMI/BMI z-score at baseline (overweight vs obese), the intervention setting (tertiary centre vs other)". The current study only presents subgroup results for intervention modality. These results are reported even though, again, substantial heterogeneity for the subgroup analyses is acknowledged. There is no mention of the results relating to the other subgroup analyses. 3. A complete qualitative narrative synthesis of studies is not provided. No details of the individual study effect sizes are presented in the text (Table 1 reports baseline BMI values only). 4. No sensitivity analyses are reported. The protocol states that analyses excluding studies found to be at high risk for bias, and analyses excluding studies with above 20% participant attrition would be carried out.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2

Dr. Malgorzata Kowal, Univ Sch Phys Educ Krakow

Comments to the Author:

1. After the corrections, which were made by the Authors of manuscript, I have no comments and recommend the article for publication.

Response: We thank the Reviewer for acknowledging the corrections that were made to the manuscript, and for recommending our article for publication.

Reviewer: 3

Ms. Julie Morris, South Manchester NHS Trust

Comments to the Author:

This systematic review has been revised to incorporate an extended search period and has increased to nine the number of trials included in the study.
However, concerns remain relating to the presentation of meta-analysis results in the presence of significant heterogeneity.

Response: We appreciate the concern from the Reviewer regarding the presentation of meta-analytic results in the presence of substantial heterogeneity. We have revised the manuscript such that the results are reported and contextualized in light of this heterogeneity. We hope these revisions improve the quality of the manuscript, and address the Reviewer's concern.

1. The results of a meta-analysis of all 9 included studies are reported in the Results section and in the Abstract, even though 'substantial heterogeneity' was present. The extent of such heterogeneity means that it is really inappropriate to present the summary effect size from the meta-analysis. In fact, the protocol (PROSPERO CRD42020166843) appears to be of the same opinion as it states that, "If there is substantial heterogeneity, a qualitative narrative summary will be done as opposed to a meta-analysis".

Response: We thank the reviewer for noting this issue. We did have substantial heterogeneity for the BMI z score outcome. We attempted to explain this heterogeneity through subgroup analyses based on the nature of the intervention. We examined multicomponent educational interventions vs the rest and motivational interviewing vs the rest. The test for subgroup effect was not significant in any of these analyses. We further attempted a leave one out approach. No single study was responsible for the heterogeneity. To address the persisting heterogeneity, we used a random-effects model which incorporates the heterogeneity and therefore the results pertain to the mean effect across studies and should be interpreted alongside the confidence intervals (which are wider for random effects models). We are comfortable presenting a summary estimate for the following reasons:

1. There is substantial overlap in the confidence intervals.
2. 9 out of 12 experimental groups showed that pediatric weight management interventions reduced BMI z score, i.e., similar direction of effect.

As such we believe a summary estimate is
reasonable. However, considering the Reviewer's concerns we have downgraded the level of evidence based on this inconsistency and further advise caution in the interpretation of these results. This approach is consistent with the guidance from the Cochrane handbook.[1]

We also acknowledge the Reviewer's concern regarding discrepancies between the protocol and our manuscript. We have included a subsection in the Discussion titled, "Deviations from

protocol" where we transparently acknowledge these deviations from the protocol (P.14 lines 334-341). It reads as follows:

"We were unable to perform subgroup analyses based on sex or differences in BMI z score at baseline due to insufficient reporting of data. Similarly, descriptions of outpatient intervention settings were vague and therefore were not compared to other settings (e.g., Primary care clinics). In the protocol we planned to report a narrative synthesis in the presence of substantial heterogeneity. However, given the direction of effect, and the use of random effects models to incorporate heterogeneity, we have reported a summary measure, but do advise caution when interpreting these results."

2. In terms of subgroup analysis, the protocol states that, "The following characteristics are thought to introduce clinical heterogeneity, and thus the following subgroup analyses are planned should the data permit: variation in sex (males vs female), differences in BMI/BMI z-score at baseline (overweight vs obese), the intervention setting (tertiary centre vs other)". The current study only presents subgroup results for intervention modality. These results are reported even though, again, substantial heterogeneity for the subgroup analyses is acknowledged. There is no mention of the results relating to the other subgroup analyses.

Response: We thank the reviewer for raising this issue and giving us the opportunity to explain our reasoning. Firstly, we agree with the Reviewer that there is no mention of the proposed subgroup analyses from our PROSPERO protocol. We have added this information in the Methods (Synthesis of results) on P.9 lines 181-189: "If there was substantial heterogeneity, subgroup analyses based on sex, differences in BMI z score at baseline (e.g., overweight vs. obesity), or intervention setting were to be performed. If substantial heterogeneity remained, a qualitative synthesis was conducted.

Sensitivity analyses excluding trials with high risk of bias, or with high rate of participant attrition (>20%) were to be performed (PROSPERO CRD42020166843)."

We have added a section (Deviations from protocol) to the manuscript (Discussion) which outlines all deviations from the protocol and an explanation for these deviations (P.14 lines 334-341): "We were unable to perform subgroup analyses based on sex or differences in BMI z score at baseline due to insufficient reporting of data. Similarly, descriptions of outpatient intervention settings were vague and therefore were not compared to other settings (e.g., Primary care clinics). In the protocol we planned to report a narrative synthesis in the presence of substantial heterogeneity. However, given the direction of effect, and the use of random effects models to incorporate heterogeneity, we have reported a summary measure, but do advise caution when interpreting these results."

We have also rewritten the sections on subgroup analyses (results) and sensitivity analysis (results) so that our approach is clear.

Subgroup analyses (P.11 lines 233-241): "An exploratory subgroup analysis was completed based on intervention modality (multicomponent educational interventions vs. MI) for BMI z score. We saw no subgroup effects when we compared the studies that used MI (MD -0.02, 95% CI -0.11, 0.06; 3 trials, 987 participants; p=0.59; I^2 49%) compared to the others (Figure 5), or when we compared multicomponent educational interventions (MD -0.11, 95% CI -0.12 to -0.09; 6 trials, 504 participants; p<0.001; I^2 68%) to the others (Figure 6). None of these subgroup analyses explained the heterogeneity. None of the preplanned subgroup analyses were possible due to missing or insufficient reporting of data."

Sensitivity analysis (P.11 lines 242-247): “All the included studies were at high risk of bias and therefore conducting sensitivity analysis based on overall risk of bias was not possible. However, the results from BMI z score in the studies at low risk of bias due to attrition had a similar direction of effect (reducing BMI z score) (i.e., attrition <20%; mean difference (MD) –0.10, 95% CI -0.11 to -0.09; 4 trials, 704 participants; p=<0.001; I^2 67%), despite substantial heterogeneity (Figure 7).”

3. A complete qualitative narrative synthesis of studies is not provided. No details of the individual study effect sizes are presented in the text (Table 1 reports baseline BMI values only).

Response: We include in the Results a subsection titled “Summary of included studies” (P.10 lines 209-222). Rather than provide a complete qualitative narrative synthesis, we described individual characteristics, including intervention duration, and methodological differences, such as risk of bias, as contributing factors to substantial heterogeneity.

We have added individual study effect sizes in Table 1.

4. No sensitivity analyses are reported. The protocol states that analyses excluding studies found to be at high risk for bias, and analyses excluding studies with above 20% participant attrition would be carried out.

Response: Thank you for raising this point. All the included studies were at high risk of bias and therefore conducting a sensitivity analysis based on overall risk of bias was not possible. However, we do include a sensitivity analysis for risk of bias due to attrition (see P.11 lines 242-247, Figure 7).

References

1. https://training.cochrane.org/handbook/archive/v6.1/chapter-10#_Ref180060218